HR Analytics Employee Attrition Analysis

1. Project Title

HR Analytics Employee Attrition Analysis Using SQL, Python, and Power BI

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2. Project Overview

Employee attrition is one of the major challenges faced by organizations. High employee turnover can increase recruitment costs, reduce productivity, and negatively impact business performance.

This project aims to analyze employee attrition data to identify key factors contributing to employee turnover. Using SQL, Python, and Power BI, the project provides meaningful insights into employee demographics, job satisfaction, salary distribution, departmental attrition, and workforce trends.

The final outcome is an interactive dashboard that helps HR professionals make data-driven decisions to improve employee retention and workforce management.

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3. Problem Statement

Organizations often struggle to understand why employees leave the company. Identifying attrition patterns and the factors influencing employee turnover is essential for improving employee satisfaction and retention.

The goal of this project is to analyze employee data and discover insights that help reduce attrition rates and improve organizational performance.

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4. Objectives

- Analyze employee attrition trends.
- Identify departments with high attrition rates.
- Understand the impact of salary on attrition.
- Analyze employee demographics.
- Evaluate job satisfaction levels.
- Identify high-risk employee groups.
- Create an interactive HR dashboard.
- Generate actionable HR recommendations.

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5. Tools and Technologies Used

Database

- MySQL

Programming Language

- Python

Python Libraries

- Pandas

- NumPy
- Matplotlib
- Seaborn

Visualization Tool

- Power BI

Query Language

- SQL

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6. Dataset Information

Dataset Name

IBM HR Analytics Employee Attrition Dataset

Total Records

1470 Employees

Important Columns

- Age
- Attrition
- Business Travel
- Department
- Distance From Home
- Education
- Education Field
- Employee Number
- Environment Satisfaction
- Gender
- Hourly Rate
- Job Involvement
- Job Level
- Job Role
- Job Satisfaction
- Marital Status
- Monthly Income
- Over Time
- Performance Rating
- Relationship Satisfaction
- Stock Option Level

- Total Working Years
- Training Times Last Year
- Work Life Balance
- Years At Company
- Years In Current Role

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7. Project Workflow

Step 1

Import Employee Attrition Dataset into MySQL

Step 2

Perform SQL-based business analysis

Step 3

Conduct Exploratory Data Analysis using Python

Step 4

Create Power BI Dashboard

Step 5

Generate HR Insights and Recommendations

Step 6

Prepare Project Report and Documentation

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8. SQL Analysis

SQL was used to perform data exploration and business analysis on employee attrition data.

Key SQL Queries Performed

Employee Overview

- Total Employees
- Total Attrition Count
- Attrition Rate
- Average Employee Salary

Department Analysis

- Department-wise Employee Count
- Department-wise Attrition Analysis
- Department-wise Average Salary

Job Role Analysis

- Job Role Attrition Count

- Job Role Employee Distribution
- Highest Attrition Job Roles

Salary Analysis

- Average Monthly Income
- Income Distribution
- High Salary vs Low Salary Attrition

Demographic Analysis

- Gender Distribution
- Marital Status Analysis
- Age Group Analysis
- Education Field Analysis

Satisfaction Analysis

- Job Satisfaction Distribution
- Environment Satisfaction Analysis
- Work-Life Balance Analysis

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9. Python Analysis

Python was used for data cleaning, preprocessing, and exploratory data analysis.

Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Data Cleaning Activities

- Checked missing values
- Verified data types
- Removed duplicate records
- Performed data validation
- Created age group categories

Exploratory Data Analysis

Employee Distribution Analysis

Analyzed workforce distribution across different departments and job roles.

Attrition Analysis

Identified employees who left the organization and analyzed their characteristics.

Salary Analysis

Studied the relationship between salary and employee attrition.

Demographic Analysis

Analyzed age, gender, education, and marital status distributions.

Satisfaction Analysis

Evaluated employee satisfaction and work-life balance metrics.

Python Visualizations

- Attrition Distribution
- Age Distribution
- Salary Distribution
- Department-wise Attrition
- Job Role Analysis
- Gender Distribution
- Correlation Heatmap
- Monthly Income Analysis

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10. Dashboard Development

Power BI was used to build an interactive HR Analytics Dashboard.

Dashboard Features

- Interactive Filters
- Dynamic KPI Cards
- Department Analysis
- Job Role Analysis
- Employee Demographics
- Attrition Insights

Slicers Used

- Department
- Gender
- Job Role
- Marital Status
- Education Field

KPI Cards

Total Employees

Displays the total workforce count.

Value: 1470

Attrition Count

Displays the number of employees who left.

Value: 237

Attrition Rate

Displays the percentage of employee turnover.

Value: 16%

Average Salary

Displays average monthly income.

Value: 6.5K

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11. Dashboard Visualizations

Job Role Attrition Analysis

Shows attrition count across different job roles.

Purpose:

- Identify job roles with high employee turnover.

Gender Distribution

Displays workforce distribution by gender.

Purpose:

- Analyze gender diversity.

Department Wise Attrition

Shows attrition count across departments.

Purpose:

- Identify departments with higher employee turnover.

Education Field Analysis

Treemap visualization displaying employee distribution across education backgrounds.

Purpose:

- Understand workforce educational composition.

Income vs Attrition Analysis

Scatter chart showing relationship between salary and attrition.

Purpose:

- Analyze impact of compensation on employee turnover.

Age Group Analysis

Displays employee distribution across age groups.

Purpose:

- Identify age segments contributing to attrition.

Job Satisfaction Analysis

Shows satisfaction levels among employees.

Purpose:

- Evaluate employee engagement and satisfaction.

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12. DAX Measures Used

The following DAX measures were created in Power BI to support dashboard calculations.

Total Employees

Total Employees = COUNT('WA_Fn-UseC_-HR-Employee-Attrition'[EmployeeNumber])

Purpose:
Calculates the total number of employees.

Attrition Count

```
Attrition Count =
CALCULATE(
   COUNT('WA_Fn-UseC_-HR-Employee-Attrition'[Attrition]),
   'WA_Fn-UseC_-HR-Employee-Attrition'[Attrition] = "Yes"
)
```

Purpose:
Calculates the number of employees who left the organization.

Attrition Rate

```
Attrition Rate =
DIVIDE([Attrition Count],[Total Employees],0)
```

Purpose:
Calculates employee turnover percentage.

Average Salary

```
Average Salary =
AVERAGE('WA_Fn-UseC_-HR-Employee-Attrition'[MonthlyIncome])
```

Purpose:
Calculates average monthly income.

Age Group Column

```
Age Group =
SWITCH(
TRUE(),
[Age] < 30,"Under 30",
[Age] <= 40,"30-40",
[Age] <= 50,"41-50",
"50+"
)
```

Purpose:
Categorizes employees into age groups.

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13. Key Insights

Employee Overview

- Total Employees: 1470
- Employees Left: 237
- Attrition Rate: 16%
- Average Monthly Salary: 6.5K

Department Analysis

- Research & Development recorded the highest attrition.
- Sales department showed the second-highest employee turnover.
- Human Resources had the lowest attrition count.

Job Role Analysis

- Laboratory Technicians experienced the highest attrition.
- Sales Executives showed significant employee turnover.
- Research Scientists were among the most affected job roles.

Gender Analysis

- Male employees represented approximately 60% of the workforce.
- Female employees represented approximately 40% of the workforce.

Age Group Analysis

- Employees below 40 years contributed most to attrition.
- Younger employees were more likely to leave compared to senior employees.

Salary Analysis

- Employees with lower monthly income showed higher attrition.
- Higher-paid employees generally demonstrated better retention.

Job Satisfaction Analysis

- Employees with lower job satisfaction levels showed greater turnover tendencies.
- Employee satisfaction appears to be strongly related to retention.

Education Analysis

- Life Sciences employees represented the largest educational group.
- Medical and Technical Degree backgrounds formed significant portions of the workforce.

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14. Business Recommendations

Improve Employee Retention

Develop targeted retention programs for high-risk departments and job roles.

Review Compensation Structure

Provide competitive salary packages and performance-based incentives.

Strengthen Employee Engagement

Conduct employee engagement programs and regular feedback sessions.

Enhance Career Development

Provide training opportunities and career growth pathways.

Improve Work-Life Balance

Promote flexible work policies and employee wellness initiatives.

Focus on High Attrition Departments

Special attention should be given to:

- Research & Development
- Sales Department

to reduce employee turnover.

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15. Conclusion

The HR Analytics Employee Attrition Analysis project successfully identified key factors influencing employee turnover using SQL, Python, and Power BI.

Through data analysis and visualization, the project provided valuable insights into employee demographics, departmental performance, salary distribution, and job satisfaction trends.

The findings highlight that attrition is significantly influenced by factors such as salary levels, job role, department, age group, and employee satisfaction.

The interactive dashboard enables HR professionals and business leaders to monitor workforce trends, identify potential risks, and make informed decisions to improve employee retention and organizational performance.

This project demonstrates the practical application of data analytics techniques in solving real-world HR challenges and supporting strategic workforce planning.